Consider a database system with following schemes:

Restaurant(Rname, Rlocation, Rfund)

Cook(Cname, Cspeciality)

Worksat(Cname, Rname, Workinghrs, Shift)

Food(Fname, Cname, Catagory)

Now write SQL statements and relational algebra statements for following Queries:

a) Select the name and location of all restaurants.

b) Find the working hours of cook named "Sita".

c) Select name of the foods cooked by "Ramesh".

d) Use join to select name of restaurants where food of category "breakfast" is available.

e) Find the name of cooks who work as "KFC"

Discuss the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples.

Design an ER diagram for Hospital System. Use your assumption for the selection attributes, entities and relationships. Show the use of total and partial participation along with the appropriate cardinalities.

Define stored procedures and triggers. How they are applicable in database? Illustrate with examples, how can you create before and after triggers on INSERT query.

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a) Design an ER diagram of your choice. The ER diagram should contain at least five entities. Out of those entities, one should be weak entity. Your entities should have appropriate attributes including primary key attributes, if any. There should be presence of one to one relationship, one to many relationship and many to many relationships in your ER diagram.

b) Convert the ER diagram in Q. 19. a. into equivalent relational tables.

Given the schema as below, write relational algebra and SQL queries for following:

Sailors(sid, sname, rating, age)Boats(bid, bname, colour)

Reserves(sid, bid, reservedate)

a) Find name, rating and age of all the sailors.

b) Find the name and color of boats having color blue.

c) Find name of the boats reserved on date 2021/07/20.

d) Find the names of boats and sailors who have reserved boats having color green.

e) Use Left and Right Outer Join between Sailors and Reserves.

Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples.

Differentiate between stored procedure and trigger. Why concurrency control is needed? Describe timestamp based concurrency protocol.

Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples.

Consider the following tables:

Book(Bookid, Title, Publisher, Published Date, Price)

Book_Authors (Bookid, Author name)

Borrower (Card No, Name, Address, Contact)

Bookissue (BookId, CardNo, IssuedDate, Due Date)Write SQL query for the following:

a) Find the name of the books taken by Ramesh Sharma.

b) Find the Issued book name and the borrower's name who have Issued date before 15th May, 2023 and due date after 15th June, 2023.c) Find the borrower's name who has taken the book written by Laxmi Prashad Devkota.

d) Find the number of books Issued on the date 25th April, 2023.

How different join operations can be performed? Explain with proper examples. Consider a database of a shipment company. Shipped items are received into the system at a single retail center. Shipped items make their way to their destination via one or more standard transportation events (i.e. flights, truck deliveries). The transportation events follow a regular schedule. Create an ER diagram to represent this information.

What are fully and partial functional dependencies? Explain 2NF and 3NF in detail.

Consider the following employee database, where primary keys are underlined Supplier (supplier-id, supplier-name, city) Suppliers (Supplier-id, parts-id, quantity) Parts (part-id, par-name, color, weight)

a) Write a SQL query to create above table.

b) Find the name of all suppliers who has supplied parts with quantity more than100 and the name of all parts supplied by "RD Traders"

c) Find the number of parts supplied by "S02"

d) Find the name of suppliers located in city "KTM".

How participation constraints are used in ER diagram? Explain. Also Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period and has an associated due date, and the date when the payment was received.

Why normalization is required in database? Explain the 1NF, 2NF and 3NF of normalization with example.

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a) What is inner join? Perform inner join to find the customer names who are going on flight to 'South Wales':

Customer (Cid, Name, Address, Age, Email)

Booking(Cid(FK), FlightID(FK), BookedDate)

Flight(FlightID, AirplaneCode, DepartureTime, Destination, Duration)

b) What are unary and ternary relationship? Write with examples.

How ER model can be mapped to the relational data model? Explain with a proper example.

What are the advantages of normalization? Explain 3NF and BCNF with example.

****BEST OF LUCK HEHEHEHE****